Solving Quadratic Equations — Diagnosing Incomplete Root Sets and Formula Sign Errors

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Most wrong answers on this topic are not arithmetic slips. They are one of two habits:

- **An incomplete root set** — a square root taken without $\pm$, a factor divided away, or only the "nice" solution reported. A quadratic has two solutions unless the discriminant is zero.
- **A sign error in the formula** — $-b$ written as b when b is already negative, or $-4ac$ subtracted when c is negative and it should be added.
- In every problem, work in this order: name the mistake in words, then show the corrected work, then state the complete solution set.

1. Warm-up. Solve $x^2 = 49$.

Toby answers $x = 7$. Is his solution set complete? Write the complete set.

Answer: ____________________

2. Find and fix. Jae solves $x^2 = 3x$ by dividing both sides by x, and reports the single solution $x = 3$.

Dividing by x assumes something about x that the equation never promised. Say what that assumption is, then solve the equation correctly by factoring and give both solutions.

Answer: ____________________

3. Find and fix. Rio computes the discriminant of $3x^2 - 7x - 6 = 0$ like this:

$$b^2 - 4ac = (-7)^2 - 4(3)(6) = -23, \quad \text{so “no real solutions”}.$$

Rio's substitution changed one of the three coefficients. Name which one and how, then compute the correct discriminant.

Answer: ________________

4. Find and fix. Mia uses the quadratic formula on $2x^2 - 4x - 3 = 0$. She reasons: “b is already negative, so the numerator starts with -4,” and reports the larger root as $x = \dfrac{-4 + \sqrt{b^2 - 4ac}}{2(2)} \approx 0.58$.

State the rule Mia broke in one sentence, then compute the correct larger root. Leave it in exact form.

Answer: ________________

5. Find and fix. Sam solves $(x-3)^2 = 25$ by taking the square root of both sides. Because he takes only the positive square root, he reports $x = 8$ as the answer.

Which solution did Sam lose, and what should he have written on the line where he took the square root? Give the complete solution set.

Answer: ________________

6. Find and fix. Kai solves $(2x+3)(x-4) = 0$ by reading the numbers straight out of the parentheses and answering $x = 3$ and $x = -4$.

Kai made two different mistakes at once — a sign and an unfinished step. Describe both, then give the correct solution set.

Answer: ________________

7. Find and fix. A student solves $4x^2 - 12x + 9 = 0$ and reports *two* solutions, $x = \frac{3}{2}$ and $x = -\frac{3}{2}$, reasoning that "every quadratic has two roots, one positive and one negative".

(a) Compute the discriminant $b^2 - 4ac$ for this equation.

(b) Use its value to say how many distinct real solutions there are, and give the correct solution set. Explain what the student's rule gets wrong.

Answer: ________________

8. Explain.

(a) Explain why taking the square root of both sides of an equation loses a solution unless you write $\pm$. Use the fact that two different numbers can have the same square.

(b) Point to where the same $\pm$ lives inside the quadratic formula, and say what it produces.

(c) Write one checking habit that would catch a missing root before you hand the work in.